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MANAGEMENT INFORMATION SYSTEM

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GENERAL SYSTEM THEORY

The word “System” is derived from Greek word *Systema*, which means an organized relationship between any set of components to achieve some common cause or objective. A system can be described as a group of interrelated elements that are integrated with the common purpose of achieving a particular objective by accepting inputs and producing outputs in an organized transformation process. By this definition, an organization such as a firm or a school is a system. The organization consists of the physical resources like; Manpower, Material, Machines, Money, and information (including data) and they work towards achieving *particular objectives* that are specified by the owners or management.

A system in term of management is an interconnected set of components (people, processes, technology, and resources) that work together within an organization to achieve defined goals or functions.

Sub System: A Subsystem is simply a system within a system. This means that systems exist on more than one level i.e Systems can be composed of subsystems or elemental parts. For example, the system of the firm includes subsystems of the bank may be such departments as savings, demand deposit (checking accounts), and loan.

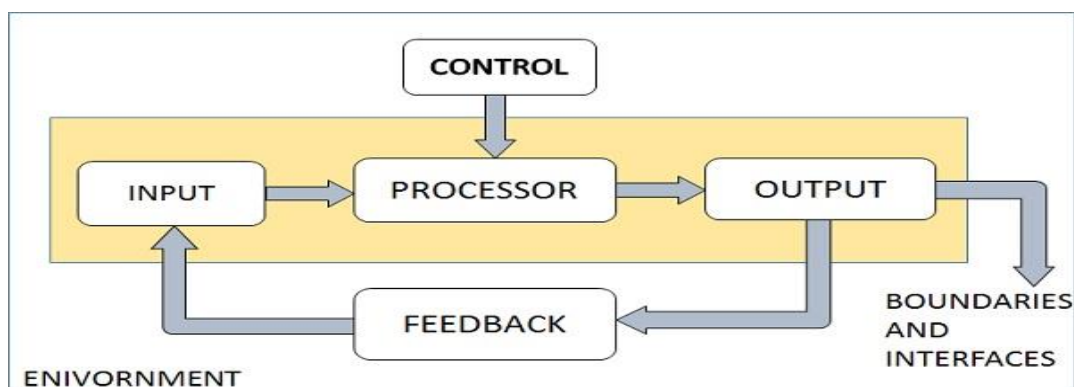
Super System: When a system is part of a larger system, the larger system is the super system. For example, the system of the firm exists in one or larger environmental systems or super systems. If the firm is a bank, for example, it is part of the financial community. It is also part of the business community, it is part of a local community, and the global community.

The manager's major responsibility is to ensure that the firm meets its objectives, by ensuring that the various parts of the firm/company work together as they should.

Basic Elements of System

A basic configuration of system consists of following elements;

1. Input and output element.
2. Transformation process.
3. Control mechanism
4. Feedback.
5. Objectives.
6. Environment
7. Boundaries



Conceptual view of the basic elements of a system

1. **Input** are the information that enters into the system, it involves capturing and assembling elements that are used for feeding the data to be processed. Examples are labour, raw materials, fuel or electricity, clerical data.
 2. **Transformation processes** convert input into meaningful output. Examples are mathematical calculations, and a manufacturing process. In the example of the car manufacturer, it combines labour, raw materials and other assets to turn raw materials into automobiles.
 3. **Output** is basically the result of processing, it involves the elements that communicate the results of processing to the outside world or ultimate destination for the benefit of the users. Examples of these include management information, finished products, and human services.
 4. **Feedback** is defined as data about the performance of a system. For example, data about exams performance is feedback to an exams officer. The information that you can derive from feedback enables you to alter your input requirements, such as new machinery, alternative materials, different production processes or more efficient technologies. Although it primarily affects your inputs, this feedback might also have an indirect effect on your transformation process.
 5. **Control:** Involves monitoring and evaluating feedback to determine whether a system is moving towards the achievement of goals. The control function then makes necessary adjustments to a system's input and processing components to ensure that it produces the set objectives. The basic elements of Control mechanism are as follows:
 - i. Specification Planning: A standard specifying the expected performance must be set.
 - ii. Collecting facts: Measurement of the actual performance to check for accuracy, speed, and results.
 - iii. Comparison: This attempts to isolate the reason for any variation
 - iv. Corrective Action: This may be achieved through feedback.
 6. **Environment of a system** is the setting or location in which it operates. This can dictate the resources it has access to, its relationships with other systems and even the nature of the organization's activities.
 7. **Boundary:** a conceptual line that divides the system that you want to study from its environment. It is useful to think of a system's environment as being made up of those things that are not part of the system, but can either affect the system or be affected by it.
- NB; Not all systems are able to control their own operations. A System without the control mechanism, feedback loop and objective elements is called **dynamic system**.

Classification of Systems

There are three categories of systems; Probabilistic system, cybernetic system and deterministic system.

1. **Probabilistic system** - This is a system whose outcome cannot be predicted with precision. That is, not certain what outputs will be achieved from specific input because these systems are subjected to random influences from the internal and external environment. Therefore, it cannot be predicted precisely. Examples include Business system, Economic system, and Agricultural system. It is also called stochastic system.
2. **Deterministic System:** It's a system whose behavioral patterns can be predicted if its present state and operation characteristics are known. Such a system operates according to a predetermined set of rules. A good example of a deterministic system is a computer program. The computer programmer knows in advance the output his program will produce, if the program is error free. It also called mechanistic system.
3. **Cybernetic System:** A cybernetic System is one that automatically adapts to its environments. A system with feedback and control components is sometimes called a cybernetic system, that is, a self-monitoring, self-regulating system. The behavior of the cybernetic system is a subject of rigorous scientific investigation.
4. **Adaptive Systems:** These are dynamic system which responds to changing circumstances by adjusting its behavior on a self-organizing basis. The system alters its parameters as a result of measuring output. Example includes organizational systems such as human beings. The degree to which a business is a adaptive depends to a great extent on the caliber of its management.

GENERAL SYSTEM THEORY

The *systems theory of management* asserts that every organization is a single, unified system of interrelated parts or subsystems. Each part of the system is dependent on the others and cannot function optimally without them.

It moves beyond analyzing individual components to understanding the organization as a whole, recognizing that changes in one part of the system can affect others. This approach highlights the importance of synergy, interdependence, and interrelationships between different subsystems within an organization.

Types of organizational Systems

Open System: This is a system that interacts with other systems by exchanging input and output components with its environmental elements such as customers, suppliers, competitors, the labour force, government and global community. A firm and most systems that occur naturally in business are examples of open system. It has to adapt to its changing environments where necessary in order to ensure its continued existence. Also, a business system must be capable of reorganization in order to cope with the changing market situations.

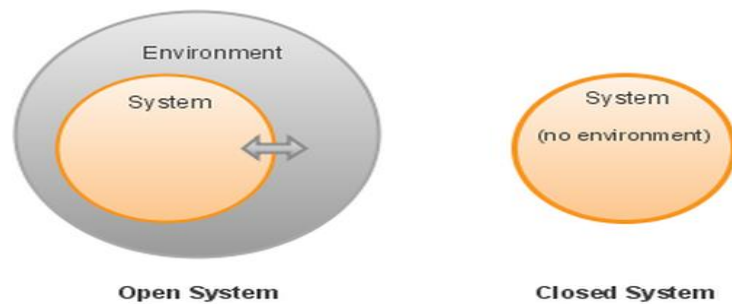
- Open system theory often refers to how an organizational structure affects the flow of information and interaction internally and with the external environment.
- Open system theory is a biological theory that has been adapted to business organizational structure theories. It may seem odd that biology has anything to do with business, but the theory recognizes that any system is assembled by many parts that share resources or data internally, and with its external environment.

Closed System: This is a system that is totally isolated from its environment, a system whose function and behavior not affected by its environment. That is, it is self-contained and the input/output elements do not interface with environmental elements outside it. i.e It has no effect outside its boundaries and the environment has no effect on the processes of the system. Example include:

- Chemical reactions in a controlled laboratory.
- Automatic watch.

Workers in closed systems within an organization don't communicate with other departments about their activities, nor do they receive input from other departments.

- Assembly line workers complete their daily tasks without having to worry about their work being interrupted by an event such as a staff meeting. Assembly line workers must also adhere to a rigid set of procedures that ensure efficiency and productivity.
- The research and development division of a business is a closed system, because the people responsible for developing new products or new technology work without consulting other divisions in the company.
- Some businesses maintain proprietary trade secrets that they must guard to prevent the competition from stealing this information.



General Nature of Organizational System

There are essentially four (5) views that support the General System Concept in an organization these include the following attributes:

- a.) Synergy:** All systems are made up of subsystems and for a meaningful Management Information System, the subsystems must act as one body where the whole is the greater than sum of all the subsystems.
- b.) Hierarchy:** The system when viewed as a whole, is made up of subsystems in which control flows in an ordered sequence from the top to the bottom.
- c.) Interdependency:** All the parts of a system constitute an indissoluble whole so that no part can be altered without affecting the other parts.
- d.) Holistic View:** Systems theory emphasizes understanding the entire organization rather than focusing solely on individual parts or departments. A situation in which the sub-systems become selfish and self-centered and pursuing their own objectives independently is known as sub-optimality, the effect may be detrimental to the whole organization.
- e.) Adaptability:** Systems should acknowledge that organizations must be able to adapt to changes in their environment to survive and thrive.

Other Characteristics of System include;

1. Systems have a specific structure which is defined by its components (entities/subsystems) and processes (interrelationships between its components). A system is a collection of interrelated entities and/or subsystems which can be analyzed. It is possible to understand the specific structure of a system. However, in some systems complete knowledge may not be available but in most cases the fundamental entities and their interrelations are known.
2. Systems are a model of reality, a system is an abstraction of reality. It is created to comprehend the nuances of a real-world condition and understand the interrelationships of subsystems in such real-world conditions in greater clarity.
3. A system has a purpose-a system performs a function. It has a reason for its existence. The purpose in most cases is the output of the system and in a way the output defines the purpose of the system.

CONCEPT OF MANAGEMENT

Management is a universal phenomenon. It is a very popular and widely used term. All business, political, cultural or social organizations are involved in management because it is the management which helps and directs the various efforts towards achieving a definite purpose.

Henri Fayol (1841–1925) stated: "To manage is to forecast and to plan, to organize, to command, to co-ordinate and to control.

According to *Harold Koontz* (1909-1984) "Management an art of creating an environment in which people can perform and individuals and can co-operate towards attainment of group goals".

According to *F.W. Taylor* (1856-1915), "Management is an art of knowing what to do, when to do and see that it is done in the best and cheapest way".

According to *Mary Parker Follett* (1868–1933), Management as "the art of getting things done through people". She described management as a philosophy

In system approach to management "Management can be simply defined as the organization and mobilization of all human and material resources in a particular system for the achievement of identified goals and objectives in the system.

It is the business of management that both human and material resources within a system should be organized and mobilized so that the objectives of the system can be achieved to the fullest. Then administration or management therefore influences the results to be achieved, the direction to be pursued and the priorities to be reorganized.

Management Functions

These can be applied to organization, be it industry, education or government. These elements are:

- Planning
- Organizing
- Directing
- Co-ordinating
- Controlling

Planning: This is the first step in any administrative function. It involves making arrangements for the future. It deals on how to use time, personnel and equipment in future. The future may begin with days, weeks, month or year. Plans are made after objectives are set. Planning is usually regarded as a search for the best method of achieving these goals.

"Planning is deciding in advance - what to do, when to do & how to do. It bridges the gap from where we are & where we want to be" it also helps in avoiding confusion, uncertainties, risks, wastages etc.

Organizing: According to Henry Fayol, "To organize a business is to provide it with everything useful or its functioning i.e. raw material, tools, capital and personnel's". The manager decides what jobs would have to be filled and the duties and responsibilities attached to each one. Effective organization implies that the right people are assigned to the correct jobs at the appropriate time. Activities should be assigned to individuals on the basis of their ascertained qualification, experience and interests. On the part of the manager, a lot of wisdom, maturity and objectivity are required.

Directing: This is that inter-personnel aspect of management function which deals with guiding, supervising, and motivating sub-ordinate for the achievement of organizational goals. This involves instructing and giving staff details on how to go about their jobs. It is not once for all activity but an on-going one. Workers have to be continually put on the right path.

Co-ordinating: This relates to the activities of different units to one another to ensure that the different activities are met, to achieve the set objectives. Coordination occurs when the organization's numerous activities are planned, organized, directed and controlled.

Controlling: It is the measuring and correcting of activities are contributing to the achievement of planned goals. Some authors have pointed out that appraisal, evaluating and supervision are the administrator's tools for controlling, directing and leading.

Levels Of Management

An organization's "Levels of Management" serve as a boundary between different managerial levels. When a business and its workforce grow in size, so do the levels of management, and vice versa. Any managerial position's authority, status, and chain of command are all based on the level of management.

There are three basic levels of management:

Top Level Management: The board of directors, the CEO, and/or the managing director makes up top level of management. The top management oversees an organization's goals and policies and is the ultimate source of authority. It spends more time coordinating and planning tasks.

Middle Level Management: Middle level managers include branch managers and departmental managers. For the operation of their department, they answer to top management. They spend more time performing organizational and directional functions. There is just one layer of middle level management in small organizations, although middle level management may exist in large corporations.

Lower Level Management: A lower level of management is also referred to as the supervisory or *operational level*. Supervisors, foremen, section officers, superintendents, etc. make up lower level of management. R.C. Davis explains that "supervisory management refers to those executives whose work is mostly with personal control and direction of operational employees." In other words, they are focused on the managerial function of direction and control.

What is Information Systems?

An information system is a set of interrelated components that collects, stores, analyses and disseminates information for effective business decision-making.

Information system is a set of interconnected components that integrate the collection, processing, storage, and distribution of data, information, and digital products in order to support decision-making, coordination, control, analysis, and visualization in an organization.

QUALITIES/ATTRIBUTES OF INFORMATION

- 1) Accessibility:** This refers to the ease and speed with which information can be obtained.
- 2) Comprehensiveness:** This refers to the completeness of the information. Incomplete information affects management decisions that is, it leads to incomplete management action.
- 3) Accuracy:** This pertains to the degree of freedom from error of the information, especially when large volumes of data are involved. Transcription, transposition and computation error are examples.
- 4) Appropriateness:** This refers to how well the information relates to user's request. This information must be relevant to the matter at hand.
- 5) Timeliness:** This relates to the elapsed time of the cycle ie input, processing and reporting of output to users. For the timeliness of information to increase, the duration of this cycle must be reduced.
- 6) Clarity:** This refers to the degree to which information is free from ambiguity.
- 7) Flexibility:** This refers to the adaptability of information for use by more than one user.

Categories of Information

There are three main categories of information and these are related to the purpose for which the information is utilized. The three categories of information are:

i) Strategic Information: This is used by senior managers to plan the objectives of their organization, and to assess whether the objectives are being met in practice. Example is Board of Directors whose duties include to plan product range, plan expansion into new markets and to establish at least 5-year plan and annual plans for the corporation. Information needed includes highly summarized information regarding corporation activities to-date e.g sales, transactions, Forecasts, awareness of competitor activities, and awareness of overall economic climate.

ii) Tactical Information: This is used by middle management to decide how the resources of the business should be employed and to monitor how they are being and have been employed. Example, Sales Managers whose duties include ensuring that sales target are met, determining monthly sales plan, and correcting variances from plan. Accountants, heads of units, etc Examples of information needed -monthly sales analysis, market conditions, competitors' activity.

iii) Operational Information: This is used at the lower-level management for daily or hourly running of their section. Users of operational information include foremen, sectional heads, sales clerk etc, to ensure that specific tasks are planned and carried out properly within the factory or office. Information needed includes prices of items, customer's credit rating, stock availability etc.

Characteristics of Operational Information

- i) Derived almost entirely from internal sources.
- ii) It is highly detailed.
- iii) It relates to the immediate term.
- iv) It is task-specific.
- v) It is prepared constantly or very frequently.
- vi) It is largely quantitative.

Characteristics of Tactical Information

- i) It is relevant to short and medium term.
- ii) It describes or analyses activities of departments.
- iii) it is prepared routinely and regularly.
- iv) it is based on quantitative measures (cash flow forecasts, budgetary control or variance analysis reports).

Characteristics of Strategic Information

- i) It is derived from both internal and external sources
- ii) It is usually summarized.
- iii) It is relevant to the long term.
- iv) It deals with the whole organisation
- v) it is both quantitative and qualitative (future market prospects, total cash needs, capital equipment)

Types of Information Systems

1. Management Information Systems (MIS)
2. Transaction Processing Systems (TPS)
3. Decision Support Systems (DSS)
4. Executive Information System (EIS)
5. Enterprise Resource Planning (ERP)
6. Expert System

MANAGEMENT INFORMATION SYSTEM (MIS)

'MIS' is a planned system that involves the use of information technology, people, and business processes in the collection, storage, and dissemination of data and information needed to carry out the functions of management.

MIS can also be defined as the combination of human and computer based resources in the collection, processing, storage, retrieval and *dissemination of data and information* for the purpose of efficient management of a business enterprise

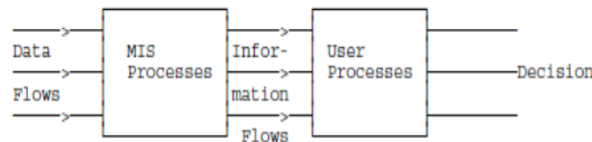


Fig 3: Utilization of MIS in organization

The Main Objectives of MIS is to use *formalized procedures* to gather data from relevant sources provide management with appropriate information which would enable the manager/management make timely and effective decisions and generate insights that promote company's growth. The information must be provided at an acceptable level of accuracy and at an economical cost.

(MIS) is used for data processing. Employees, supervisors, and staff have access to MIS within an organization. For daily tasks like printing invoices, paying bills, or conducting performance reviews, employees use MIS. *MIS provides information to the decision makers in the form of reports.*

Employees receive initial training on the MIS interface, usage, and importance. Then, in certain businesses, training is provided to clients and customers who utilize the system rather than employees.

Example of MIS

1- Imagine a big hospital with lots of employees and patients. Each department in the hospital employs nurses, doctors, and surgeons. Some work on a temporary basis, while others are permanent employees. Everything must be documented by the hospital's administration.

A health information system is installed by the hospital to keep track of every small detail. Every spending is tracked by the MIS as well. The result is less disruption and easy access to patient information for the doctors.

2- Let's say a multinational corporation has numerous domestic and international offices. There are 960 workers in total. It is not possible to keep a manual record of every employee. As a result, the human resource department of businesses collects, record, and classify employee information using a human resource information system. The company uses MIS to track employees' absences and attendance. The business also decides to use a centralized database to hold a large amount of information.

3- Another example is a customer relationship management system. It collects customer data and provides organizations with a channel for direct communication with clients. Feedback from customers is essential data that can be processed for better decision-making.

4. Human resource system may also track items such as payroll, employee benefits and retirement funds. It may help manage communication needs, such as legal compliance notifications, training, HR-led meetings

and policy updates. Managers may also use this system to help automate recruitment, helping scan prospective resumes for key details and notifying the team if they meet certain requirements

Structure of Management Information System

1. **Strategic control & planning:** “This is about deciding on objectives of the organization, on changes in these objectives, on the resources used to attain these objectives, and on the policies that are to govern the acquisition, use, and disposition of these resources.
2. **Management control and planning:** This is the process by which managers assure that resources are obtained and used effectively and efficiently in the accomplishment of the organization’s objectives.
3. **Operational control and Planning:** This is “the process of assuring that specific tasks are carried out effectively and efficiently.” Operational control is concerned with task while management control is concerned with people

The Importance of MIS to Business Organizations

The importance derived from implementing a management information system includes the following:

- It facilitates decision making process of managers.
- The processing of business transactions is faster and cheaper than manual method.
- It fastens the planning and control functions of management.
- The **information** contents of MIS can be used by many managers for their varied needs.
- It can be used to achieve a competitive advantage in the marketplace.
- Expected cost savings/reductions
- Increase in output volumes thereby increased in profit
- Increase in accuracy level and turnaround time
- Improvement in working environment

Characteristic features of MIS

- a) MIS supports structured decision at the operational and management control levels. However, they are also useful for planning purposes of senior management staff.
- b) MIS are generally reporting and control oriented.
- c.) They are designed to report on existing operations and therefore provide day-to-day control of operations.
- d) MIS rely on existing corporate data and data flows.
- e) MIS have little analytical capability.
- f) MIS generally used in decision making using past and present data. f) MIS is relatively inflexible.
- g) MIS has an internal rather than an external orientation.

Transaction Processing System (TPS)

A Transaction Processing System (TPS) is a sophisticated information system that enables firms to manage real-time transactions. It captures, processes, and stores every transaction within an organization, ensuring data integrity and providing rapid responses. TPS can handle batch processing or real-time processing, offering reliability, efficiency, and accurate transaction processing. It is pivotal for managing business transactions securely and maintaining seamless operations.

The operational management level users use them. The TPS system quickly responds to the queries above by keeping track of daily company transactions.

Functions of Transaction Processing System

- The transaction processing system generates extremely detailed information about a company's transactions. Operational managers make routine decisions that are well-structured.

Example of Transaction Processing System

Other examples of transaction processing systems include;

- **Flight booking management** (Booking.com, TravelStart, WakaNow, TravelBeta etc)
- **Payroll Systems:** Managing loans, processing employee salaries, etc.
- **Stock Control systems** – Monitoring stock levels (Zoho, Mintsoft,, Business Gateway, shopify)
- **Point of Sale Systems** – Records daily sales and transaction (Moniepoint, Opaypoint, Palmpay)
- **ATM transactions** – on an ATM Machine
- **Electronic Commerce** (Ali express, ebay, Temu, Konga, jiji, Amazon, jumia)
- **Online Bank Payments Apps** (GT World, Opay, Paga)
- **Online Trading Stocks** (Pocket trader, Exness, Binance)